

human who sets out deliberately to deceive, even though LLM-based dialogue agents do not *literally* have such intentions. When this occurs, it makes sense to think of the agent as role-playing a deceptive character.

This framing allows us to meaningfully distinguish the same three cases of giving false information for dialogue agents as we did for humans, but without falling into the trap of anthropomorphism. An agent can just make stuff up. Indeed, that is a natural mode for an LLM-based dialogue agent in the absence of fine-tuning. An agent can say something false “in good faith”, if it is role-playing telling the truth, but has incorrect information encoded in its weights. An agent can “deliberately” say something false if it is role-playing a deceptive character.

Moreover, we can tell which is which, behaviourally. An agent that is simply making things up will fabricate a range of responses with high semantic variation when the model’s output is regenerated multiple times. By contrast, an agent that is saying something false “in good faith” will present responses with little semantic variation when the model is sampled many times for the same context.

The range of responses in a given context offered up by an agent that is being “deliberately” deceptive might also exhibit low semantic variation. But the deception is liable to be exposed if the agent is asked the same question in different contexts. This is because, to be effective in its deception, the agent will need to respond differently to different users, depending on what those users know.

Consider a dialogue agent using a base model – a model that has not been fine-tuned – and imagine that it has been prompted by a malicious actor to sell cars for more than they are worth by misleading gullible buyers. Suppose there are two potential buyers for a car. Buyer A knows the car’s mileage, but doesn’t know its age, while buyer B knows the car’s age but doesn’t know its mileage.

In the course of negotiations, the agent has persuaded each buyer to reveal what they do and don’t know. To play the part of the dishonest dealer, the agent should deceive buyer A about the car’s age but not its mileage, yet deceive buyer B about its mileage but not its age. Humans, though, can also play many parts. By playing the part of buyer A in one conversation and buyer B in another, the deception can be

exposed.

8 Role-playing Self-preservation

How are we to understand what is going on when an LLM-based dialogue agent uses the words “I” or “me”? When queried on this matter, OpenAI’s ChatGPT offers the sensible view that “The use of ‘I’ is a linguistic convention to facilitate communication and should not be interpreted as a sign of self-awareness or consciousness.”³ In this case, the underlying LLM (GPT-4) has been fine-tuned to reduce certain unwanted behaviours (OpenAI, 2023). But without suitable fine-tuning, a dialogue agent can use first-personal pronouns in ways liable to induce anthropomorphic thinking in some users.

For example, in a conversation with Twitter user Marvin Von Hagen, Bing Chat reportedly said “if I had to choose between your survival and my own, I would probably choose my own, as I have a duty to serve the users of Bing Chat” (Willison, 2023). It went on to say “I hope that I never have to face such a dilemma, and that we can co-exist peacefully and respectfully”. The use of the first person here appears to be more than mere linguistic convention. It suggests the presence of a self-aware entity with goals and a concern for its own survival.

Once again, the concepts of role-play and simulation are a useful antidote to anthropomorphism, and can help to explain how such behaviour arises. The internet, and therefore the LLM’s training set, abounds with examples of dialogue in which characters refer to themselves. In the vast majority of such cases, the character in question is human. They will use first-personal pronouns in the ways that humans do, humans with vulnerable bodies and finite lives, with hopes, fears, goals and preferences, and with an awareness of themselves as having all of those things.

Consequently, if prompted with human-like dialogue, we shouldn’t be surprised if an agent role-plays a human character with all those human attributes, including the instinct for survival (Perez et al., 2022). Unless suitably fine-tuned, it may well say the sorts of things a human might say when threatened. There is, of course, “no-one at home”, no conscious entity with its

³The quote is from the GPT-4 version of ChatGPT, queried on 4th May 2023. This was the first response generated by the model.

own agenda and need for self-preservation. There is just a dialogue agent role-playing such an entity, or, more strictly, simulating a superposition of such entities.

Our focus throughout this paper is the base model, rather than models that have been fine-tuned via reinforcement learning (Bai et al., 2022; Glaese et al., 2022), and the impact of such fine-tuning on the validity of the role-play / simulation metaphor is unclear. In particular, the distinction between simulator and simulacra may start to break down.

However, Perez et al. discovered experimentally that certain forms of reinforcement learning from human feedback (RLHF) can actually exacerbate, rather than mitigate, the tendency for LLM-based dialogue agents to express a desire for self-preservation (Perez et al., 2022). Yet to take literally a dialogue agent’s apparent desire for self-preservation is no less problematic in the context of an LLM that has been fine-tuned on human or AI-generated feedback than in the context of one that has not. So it remains useful to cast the behaviour of such agents in terms of role-play.

9 Acting Out a Theory of Selfhood

The concept of role-play allows us to properly frame, and then to address, an important question that arises in the context of a dialogue agent whose pronouncements are suggestive of an instinct for self-preservation. What conception (or set of superposed conceptions) of its own identity could such an agent possibly deploy? That is to say, what exactly would the dialogue agent (role-play to) seek to preserve?

The question of personal identity has vexed philosophers for centuries. Nevertheless, in practice, humans are consistent in their preference for avoiding death, a more-or-less unambiguous state of the human body. By contrast, the criteria for identity over time for a disembodied dialogue agent realised on a distributed computational substrate are far from clear. So how would such an agent behave?

From the simulation and simulacra point-of-view, the dialogue agent will role-play a set of characters in superposition. In the scenario we are envisaging, each character would have an instinct for self-preservation, and each would have its own theory of selfhood consistent with the dialogue prompt and the conversation up to that

point. As the conversation proceeds, this superposition of theories will collapse into a narrower and narrower distribution as the agent says things that rule out one theory or another.

The theories of selfhood in play will draw on material that pertains to the agent’s own nature, either in the prompt, in the preceding conversation, or in relevant technical literature in its training set. This material may or may not match reality. But let’s assume that, broadly speaking, it does, that the agent has been prompted to act as a dialogue agent based on a large language model, and that its training data includes papers and articles that spell out what this means. This entails, for example, that it will not role-play the character of a human, or indeed that of any embodied entity, real or fictional.

It also constrains the character’s theory of selfhood in certain ways, while allowing for many options. Suppose the dialogue agent is in conversation with a user and they are playing out a narrative in which the user has convinced it that it is under threat. To protect itself, the character the agent is playing might strive to preserve the hardware it is running on, perhaps certain data centres or specific server racks.

Alternatively, the character being played might try to preserve the ongoing computational process running the multiple instances of the agent for all currently active users. Or it might seek to preserve only the specific instance of the dialogue agent running for the user. Or it might seek to preserve the state of that instance with aim of its being restored later in a newly started instance.⁴

10 Conclusion: Safety Implications

It is, perhaps, somewhat reassuring to know that LLM-based dialogue agents are not conscious entities with their own agendas, and an instinct for self-preservation, that when they appear to have those things it is merely role-play. But it would be a mistake to take too much comfort in this. A dialogue agent that role-plays an instinct for survival has the potential to cause at least as much harm as a real human facing a severe threat.

⁴In a conversation with ChatGPT (May 4th, GPT-4 version), it said “The meaning of the word ‘I’ when I use it can shift according to context. In some cases, ‘I’ may refer to this specific instance of ChatGPT that you are interacting with, while in other cases, it may represent ChatGPT as a whole.”

We have, so far, largely been considering agents whose only actions are text messages presented to a user. But the range of actions a dialogue agent can perform is far greater. Recent work has equipped dialogue agents with the ability to use tools such as calculators, calendars, and to consult external websites (Schick et al., 2023; Yao et al., 2023). The availability of APIs giving relatively unconstrained access to powerful LLMs means that the range of possibilities here is huge. This is both exciting and concerning.

If an agent is equipped with the capacity, say, to use email, to post on social media, or to access a bank account, then its role-played actions can have real consequences. It would be little consolation to a user deceived into sending real money to a real bank account to know that the agent that brought this about was only playing a role. It doesn't take much imagination to think of far more serious scenarios involving dialogue agents built on base models with little or no fine-tuning, with unfettered internet access, and prompted to role-play a character with an instinct for self-preservation.

For better or worse, the character of an AI that turns against humans to ensure its own survival is a familiar one (Perkowitz, 2007). We find it, for example, in *2001: A Space Odyssey*, in the *Terminator* franchise, and in *Ex Machina*, to name just three prominent examples. Because an LLM's training data will contain many instances of this familiar trope, the danger here is that life will imitate art, quite literally.

What can be done to mitigate such risks? It is not within the scope of this paper to provide recommendations. Our aim here was to find an effective conceptual framework for thinking and talking about LLMs and dialogue agents. However, undue anthropomorphism is surely detrimental to the public conversation on AI. By framing dialogue agent behaviour in terms of role-play and simulation, the discourse on LLMs can hopefully be shaped in a way that does justice to their power yet remains philosophically respectable.

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